



Ethinylestradiol 0.035mg & Cyproterone Acetate 2mg

Product Description

6.0 mm, white, round biconvex tablet, one side having 'R' logo and other side contains break line.

Pharmacodynamics

Mode of Action: Ethinyl estradiol: Estrogens are responsible for the development and maintenance of the female reproductive system and secondary sexual characteristics. Estrogens modulate the pituitary secretion of gonadotropins, luteinizing hormone, and follicle-stimulating hormone through a negative feedback system. Estrogen increases levels of sex hormone-binding globulin (SHBG) and may reduce unbound androgen levels. Ethinyl estradiol is a synthetic derivative of estradiol. The addition of the ethinyl group prevents rapid degradation by the liver.

Cyproterone: Steroidal compound with antiandrogenic, antigonadotropic and progestin-like activity

Pharmacokinetics Cyproterone acetate

Following oral administration cyproterone acetate is completely absorbed in a wide dose range. The ingestion of GIANE-35 effects a maximum serum level of 15mg cyproterone acetate/ml at 1.6 hours. Thereafter, drug serum levels decrease in two disposition phases characterized by half-lives of 0.8 hours and 2.3 days. The total clearance of cyproterone acetate from serum was determined to 3.6 ml/min/kg. Cyproterone acetate is metabolized by various pathways including hydroxylations and conjugations. The main metabolite in human plasma is the 15 β -hydroxy derivative. Some dose parts are excreted unchanged with bile fluid. Most of the dose is excreted in form of metabolites at a urinary to biliary ratio of 3:7. The renal and biliary excretion was determined to proceed with a half-life of 1.9 days. Metabolites from plasma were eliminated at a similar rate (half-life of 1.7 days).

Cyproterone acetate is almost exclusively bound to plasma albumins. About 3.5-4.0 % of total drug levels are present unbound. Because protein binding is non-specific, changes in SHBG (sex hormone binding globulin) levels do not affect the pharmacokinetics of cyproterone acetate.

Due to the long half-life of the terminal disposition phase from plasma (serum) and the daily intake, cyproterone acetate accumulates during one treatment cycle. Mean maximum drug serum levels increased from 15 ng/ml (day 1) to 21 ng/ml and 24 ng/ml, at the end of treatment cycles 1 and 3, respectively. The area under the concentration versus time profile increased 2.2 fold (end of cycle 1) and 2.4 fold (end of cycle 3). Steady-state conditions were reached after about 10 days. During long-term treatment cyproterone acetate accumulates over treatment cycles by a factor of 2.

The absolute bioavailability of cyproterone acetate is almost complete (88% of dose). The relative bioavailability of cyproterone acetate from GIANE-35 was 109 % when compared to an aqueous microcrystalline suspension.

Ethinylestradiol

Orally administered ethinylestradiol is rapidly and completely absorbed. Following ingestion of GIANE-35, maximum drug serum levels of about 80 pg/ml are reached at 1.7 hours. Thereafter, ethinylestradiol plasma levels decrease in two phases characterized by half-lives of 1 - 2 hours and about 20 hours. For analytical reasons these parameters can only be calculated for higher dosages.

For ethinylestradiol an apparent volume of distribution of about 5 l/kg and a metabolic clearance rate from plasma of about 5 ml/min/kg were determined.

Ethinylestradiol is highly but non-specifically bound to serum albumin. 2% of drug levels are at present unbound. During absorption and first- liver passage ethinylestradiol is metabolized resulting in a reduced absolute and variable oral bioavailability. Unchanged drug is not excreted. Ethinylestradiol metabolites are excreted at a urinary to biliary ratio of 4:6 with a half-life of about one day.

Due to the half-life of the terminal disposition phase from plasma and the daily ingestion, steady-state plasma levels are reached after 3 - 4 days and are higher by 30 - 40% as compared to a single dose. The relative bioavailability (reference: aqueous microcrystalline suspension) of ethinylestradiol from GIANE-35 was almost complete.

The systemic availability of ethinylestradiol might be influenced in both directions by other drugs. There is, however, no interaction with high doses of vitamin C. Ethinylestradiol induces the hepatic synthesis of SHBG (sex hormone binding globulin) and CBG (corticoid binding globulin) during continuous use. The extent of SHBG induction, however, is dependent upon the chemical structure and dose of the coadministered progestin. During treatment with GIANE-35 SHBG concentrations in serum increased from about 100 nmol/l to, 300 nmol/l and the serum concentrations of CBG increased from about 50 mcg/ml to 95 mcg/ml.

Indication

- Treatment of moderate to severe acne related to androgen-sensitivity (with or without seborrhoea) and/or hirsutism in women of reproductive age.
- For the treatment of acne, GIANE 35 TABLET should only be used after topical therapy or systemic antibiotic treatments have failed.
- Since GIANE 35 TABLET is also a hormonal contraceptive, it should not be used in combination with other hormonal contraceptives.

Recommended Dose

Note:

GIANE 35 TABLET should not be prescribed for the purpose of contraception alone. However, when taken as recommended, GIANE 35 TABLET will provide reliable contraception in patients treated for the above clinical conditions. If patient compliance is uncertain and contraception is necessary, then a supplementary non-hormonal contraceptive method should be considered.

The first tablet is taken on the 1st day of menstruation (1st day of the menstrual bleeding). 1 tab is then taken everyday until the pack is used up. After a tablet –free interval of 7 days during which a menstruation- like bleeding will occur, tablet taking is resumed with a new 21-day pack, followed by another 7-day tablet –free interval etc.

Mode of Administration

Route: Oral

Contraindication

Preparations containing oestrogen/progesterone combinations should not be used in the presence of any of the conditions listed below. Should any of the conditions appear for the first time during their use, the product should be stopped immediately.

- Thrombosis (venous or arterial) present or in history (e.g. deep venous thrombosis, pulmonary embolism, myocardial infarction, cerebrovascular accident).
- Presence or history of prodromi of a thrombosis (e.g. transient ischaemic attack, angina pectoris).
- Diabetes mellitus with vascular involvement.
- The presence of a severe or multiple risk factor(s) for venous or arterial thrombosis may also constitute a contraindication.
- Presence or history of severe hepatic disease as long as liver function values have not returned to normal.
- Presence or history of liver tumours (benign or malignant).
- Known or suspected malignant conditions of the genital organs or the breasts, if sex steroid-influenced.
- Undiagnosed vaginal bleeding.
- Known or suspected pregnancy.
- Lactation.
- Hypersensitivity to any of the components of GIANE-35

Giane 35 is contraindicated for concomitant use with the medicinal products containing ombitasvir/paritaprevir /ritonavir and dasabuvir (See Section Warnings and Precautions and Section Interactions with Other Medicaments).

Warning and Precautions

Warning:

Reasons for immediate Discontinuation of Medication : Occurrence for the first time of migrainous headaches or more frequent occurrence of unusually severe headaches; sudden perceptual disorders (eg: disturbances of vision or hearing); first symptoms of thrombophlebitis or thromboembolism; a feeling of pain and tightness in the chest; pending operations (6 weeks beforehand) and immobilization eg, following accidents. In all these cases, there may be an increased risk of thrombosis. Further reasons for discontinuation are : Onset of jaundice; onset of hepatitis; generalized pruritus; increase in epileptic seizures; significant rise in blood pressure; pregnancy.

Precaution:

Before starting treatment, a thorough general medical (including blood pressure measurement, urine test for sugar, and, if necessary, special liver tests) and gynaecological examination (including the breasts and cytological smear of the cervix) should be carried out to detect any diseases requiring treatment or any risks. Control examinations are recommended at about 6-monthly intervals during the use of Giane 35.

The family case history should be carefully noted. Disturbances of the clotting system must be ruled out if any members of the family have suffered from thromboembolic diseases (eg, deep vein thrombosis, stroke, myocardial infarction) already at a young age. Pregnancy must be excluded.

During treatment, ovulation will not take place, thus preventing a possible conception.

The simultaneous use of hormonal or other contraceptives is therefore unnecessary.

Women suffering from diabetes, hypertension, varicose veins, otosclerosis, multiple sclerosis, epilepsy, porphyria, tetany, chorea minor and women with a history of phlebitis or with a tendency to diabetes should be closely supervised.

In rare cases, benign and, in even rarer cases, malignant liver tumours leading in isolated cases to life-threatening intra-abdominal haemorrhage have been observed after the use of hormonal substances eg. those contained in Giane 35. If severe upper abdominal complaints, liver enlargement or signs of intra-abdominal haemorrhage occur, a liver tumour should be included in the differential-diagnostic considerations.

According to the present state of knowledge, an association between the use of progestogen-estrogen combinations and an increased risk of venous and arterial thromboembolic diseases cannot be ruled out.

The relative risk of arterial thromboses (eg, stroke, myocardial infarction) appears to increase further when heavy smoking, increasing age and the use of combined oral contraceptives coincide.

Giane 35 is not for use in men.

ALT elevations

During clinical trials with patients treated for hepatitis C virus infections (HCV) with the medicinal products containing ombitasvir/ paritaprevir/ritonavir and dasabuvir with/without ribavirin, transaminase (ALT) elevations higher than 5 times the upper limit of normal (ULN) occurred significantly more frequent in women

using ethinylestradiol-containing medications such as combined hormonal contraceptives (CHCs). Patients who are taking ethinylestradiol-containing medicinal products must switch to an alternative method of contraception (e.g. progestin only contraception or non-hormonal methods) prior to initiating ombitasvir/paritaprevir /ritonavir and dasabuvir therapy (See Section Contraindications and Section Interactions with Other Medicaments).

Circulatory Disorders

Epidemiological studies have suggested an association between the use of COCs and an increased risk of arterial and venous thrombotic and thromboembolic diseases eg, myocardial infarction, deep venous thrombosis, pulmonary embolism and of cerebrovascular accidents. These events occur rarely.

The risk of venous thromboembolism (VTE) is highest during the 1st year of use. The increased risk is present after initially starting a COC or restarting (following a 4-week or greater pill-free interval) the same or a different COC. Data from a large, prospective 3-armed cohort study suggest that this increased risk is mainly present during the first 3 months.

Overall, the risk for VTE in users of low estrogen dose (<50 mcg ethinylestradiol) COCs is 2- to 3-fold higher than for non-users of COCs who are not pregnant and remains lower than the risk associated with pregnancy and delivery.

Venous thromboembolism may be fatal in 1-2% of the cases. Venous thromboembolism manifesting as deep venous thrombosis (DVT) and/or pulmonary embolism may occur during the use of all COCs.

Extremely rarely, thrombosis has been reported to occur in other blood vessels eg, hepatic, mesenteric, renal, cerebral or retinal veins and arteries in COC users. There is no consensus as to whether the occurrence of these events is associated with the use of COCs.

Symptoms of DVT can include: Unilateral swelling of the leg or along a vein in the leg; pain or tenderness in the leg which may be felt only when standing or walking; increased warmth in the affected leg; red or discolored skin on the leg.

Symptoms of pulmonary embolism (PE) can include: Sudden onset of unexplained shortness of breath or rapid breathing; sudden coughing which may bring up blood; sharp chest pain which may increase with deep breathing; sense of anxiety; severe lightheadedness or dizziness; rapid or irregular heartbeat. Some of these symptoms (eg, shortness of breath, coughing) are nonspecific and might be misinterpreted as more common or less severe events (eg, respiratory tract infections).

An arterial thromboembolic event can include cerebrovascular accident, vascular occlusion or myocardial infarction (MI).

Symptoms of a cerebrovascular accident can include: Sudden numbness or weakness of the face, arm or leg, especially on 1 side of the body; sudden confusion, trouble speaking or understanding; sudden trouble seeing in one or both eyes; sudden trouble walking, dizziness, loss of balance or coordination, sudden, severe or prolonged headache with no known cause; loss of consciousness or fainting with or without seizure.

Other signs of vascular occlusion can include: Sudden pain, swelling and slight blue discoloration of an extremity; acute abdomen.

Symptoms of MI can include: Pain, discomfort, pressure, heaviness, sensation of squeezing or fullness in the chest, arm or below the breastbone; discomfort radiating to the back, jaw, throat, arm, stomach; fullness, indigestion or choking feeling; sweating, nausea, vomiting or dizziness; extreme weakness, anxiety or shortness of breath; rapid or irregular heartbeats. Arterial thromboembolic events may be fatal.

The risk of venous or arterial thrombotic/thromboembolic events, or of a cerebrovascular accident increases with: Age; obesity (body mass index >30 kg/m²); a positive family history (ie, venous or arterial thromboembolism in a sibling or parent at a relatively early age). If hereditary predisposition is known or suspected, the woman should be referred to a specialist for advice before deciding about any COC use; prolonged immobilization, major surgery, any surgery to the legs or major trauma. In these situations, it is advisable to discontinue COC use (in the case of elective surgery, at least 4 weeks in advance) and not to resume until 2 weeks after complete remobilization; smoking (with heavier smoking and increasing age, the risk further increases especially in women >35 years); dyslipoproteinemia; hypertension; migraine; valvular heart disease; atrial fibrillation.

There is no consensus about the possible role of varicose veins and superficial thrombophlebitis in venous thromboembolism.

The increased risk of thromboembolism in the puerperium must be considered (for information on pregnancy and lactation, see Use in pregnancy and Use in lactation under Contraindications).

Other medical conditions which have been associated with adverse circulatory events include diabetes mellitus, polycystic ovary syndrome, systemic lupus erythematosus, hemolytic uremic syndrome, chronic inflammatory bowel disease (Crohn's disease or ulcerative colitis) and sickle cell disease.

An increase in frequency or severity of migraine during COC use (which may be prodromal of a cerebrovascular event) may be a reason for immediate discontinuation of the COC.

Biochemical factors that may be indicative of hereditary or acquired predisposition for venous or arterial thrombosis include activated protein C (APC) resistance, hyperhomocysteinemia, antithrombin-III deficiency, protein C deficiency, protein S deficiency, antiphospholipid antibodies (anticardiolipin antibodies, lupus anticoagulant).

When considering risk/benefit, the physician should take into account that adequate treatment of a condition may reduce the associated risk of thrombosis and that the risk associated with pregnancy is higher than that associated with low-dose COCs (<0.05 mg ethinylestradiol).

Warning for Cyproterone acetate

Direct hepatic toxicity, including jaundice, hepatitis and hepatic failure, which has been fatal in some cases, has been reported in patients treated with 100mg or more of cyproterone acetate. Most reported cases are in men with prostatic cancer. Toxicity is dose-related and develops, usually, several months after treatment has begun. Liver function tests should be performed pre-treatment and whenever any symptoms or signs suggestive of hepatotoxicity occur. If hepatotoxicity is confirmed, cyproterone acetate should normally be withdrawn, unless the hepatotoxicity can be explained by another cause, e.g. metastatic disease, in which case cyproterone acetate should be continued only if the perceived benefit outweighs the risk.

Interactions with other Medicaments

Drug interactions which result in an increased clearance of sex hormones can lead to breakthrough bleeding and oral contraceptive failure. This has been established with hydantoins, barbiturates, primidone, carbamazepine and rifampicin; oxcarbamazepine, topiramate, felbamate and griseofulvin are also suspected. The mechanism of this interaction appears to be based on the hepatic enzyme-inducing properties of these drugs. Maximal enzyme induction is generally not seen for 2-3 weeks but may then be sustained for at least 4 weeks after the cessation of drug therapy.

Contraceptive failures have also been reported with antibiotics, such as ampicillins and tetracyclines. The mechanism of this effect has not been elucidated.

Women on short-term treatment with any of the above-mentioned classes of drugs or individual drugs should temporarily use a barrier method in addition to the GIANE-35, i.e. during the time of concomitant drug administration and for 7 days after their discontinuation. For women on rifampicin, a barrier method should be used in addition to the GIANE-35 during the time of rifampicin administration and for 28 days after its discontinuation. If concomitant drug administration runs beyond the end of the tablets in the GIANE-35 pack, the next GIANE-35 pack should be started without the usual tablet-free interval.

In women on long-term treatment with hepatic enzyme-inducing drugs, experts have recommended to increase the contraceptive steroid doses. If a high contraceptive dosage is not desirable or appears to be unsatisfactory or unreliable, e.g. in the case of irregular bleeding, another method of contraception should be advised.

Concomitant use with the medicinal products containing ombitasvir/paritaprevir/ ritonavir and dasabuvir, with or without ribavirin may increase the risk of ALT elevations (See Section Contraindications and Section Warnings and Precautions). Therefore, users must switch to an alternative method of contraception (e.g., progestogen-only contraception or non hormonal methods) prior to starting therapy with this combination drug regimen. GIANE-35 can be restarted 2 weeks following completion of treatment with this combination drug regimen.

Pregnancy and Lactation

Use in Pregnancy

The administration of GIANE-35 is contraindicated during pregnancy.

If pregnancy occurs during medication with GIANE-35, the preparation is to be withdrawn immediately.

Use in Lactation

The administration of GIANE-35 is also contraindicated during lactation. Cyproterone acetate is transferred into the milk of lactating women. About 0.2% of the maternal dose will reach the newborn via milk corresponding to a dose of about 1 mcg/kg. During established lactation, 0.02% of the daily maternal dose of ethinylestradiol could be transferred to the newborn via milk.

Side Effects

Serious undesirable effects of Giane-35 have been referred to in the contraindications and warnings and precautions sections.

The following undesirable effects have been reported in users of Giane-35 and the association has been neither confirmed nor refuted:

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| • breast tenderness, pain, secretion | • vomiting |
| • headache | • changes in vaginal secretion |
| • migraine | • various skin disorders |
| • changes in libido | • fluid retention |
| • depressive moods | • change in body weight |
| • contact lens intolerance | • hypersensitivity reaction. |
| • nausea | • Vascular Disorders Rare: Thromboembolism |

Symptoms And Treatment of Overdose

Symptoms that may occur in this case are nausea, vomiting and, in young girls, slight vaginal bleeding. There have been no reports of serious deleterious effects from overdose. There are no antidotes and further treatment should be symptomatic.

Storage Condition

Store in a cool place (below 30°C), protected from light and children.

Shelf Life

2 Years from the date of manufacturing.

Pack Size

1x21's Tablets.

Manufactured by :

 **RENATA LIMITED**
Section VII, Milk Vita Road
Mirpur, Dhaka, Bangladesh
Date of Revision : April, 2020

C-Code: